AI & MACHINE LEARNING ASSIGNMENTS

How to Approach Your Assignment

Each assignment describes a real-world problem from a real estate business. Your goal is to build a small working solution that demonstrates how you understand the problem and how you would solve it with software, AI/ML, automation, or marketing intelligence. You are not expected to build a production-ready enterprise system.

What Every Candidate Should Submit

- A working prototype, app, notebook, or API. A presentation alone is not enough.
- Source code and a README explaining how to install, configure, and run the solution.
- Sample input data and sample output so the evaluator can test the work.
- A short demo and a brief explanation of the approach, key decisions, and limitations.
- A note listing any AI model, API, library, or external service used.
- Use synthetic/anonymized data and never include passwords, API keys, or real customer personal information in the submission.

What We Will Evaluate

Evaluation criterion	Weight
Problem understanding and business relevance	20%
Working implementation and correctness	30%
AI/ML approach and quality of results	20%
Code quality, structure, and reliability	15%
Usability, demo, and explanation	15%

Important: You may use a simpler rule-based approach if it is justified and works well. If you use AI, explain what the AI does, what information it uses, and how you check its output.

AI Property Video Generator

Track: AI Content Creation / Multimedia

The Problem

A real estate sales team has property photos and basic details, but creating a separate promotional video for every property takes time. Build a tool that turns the supplied information into a short, ready-to-share property video.

What You Will Be Given / What to Use

Property name, location, price, property type, key features, contact/CTA text, and 4–8 property images. Use sample data if no real data is provided.

Your Task — Step by Step

- Create a simple form or command-line input for the property details and images.
- Use an AI model or clear template logic to write a short voice-over/script and captions. Only use facts supplied in the input.
- Arrange the images into scenes. Add readable text overlays, transitions, and optional narration or royalty-free background audio.
- Export an MP4 file and show a preview or provide a way to save it.
- Handle missing images or missing details gracefully, and show a useful error message instead of crashing.

What to Submit

- Runnable application or CLI
- At least one sample MP4 video
- README with setup instructions, dependencies, and any API/model used

How We Will Judge Success

A candidate can enter property details and images, then produce a coherent video without manually editing it in a video editor.

Example to Help You Understand

For example, a sample 2-BHK apartment listing could become a 20–40 second video with an opening title, 3–5 feature scenes, and a final enquiry prompt.